NATIONAL EXAMINATION REVALUATION BOARD

End-Semester University Degree Examination

Course: CS301 — Data Structures & Algorithms (Regulation 2025)

TIME: 3.00 HOURS **MAXIMUM: 100 MARKS** **PAPER CODE: CS301-QP-2025**

General Instructions: *Attempt all questions. Show complete algebraic derivations, binary split trees, and priority queue relaxation steps. Rough work must be labeled clearly.*

PART A — (1 × 35 = 35 Marks)

Question 1. [Recurrence Relations & Asymptotic Analysis] [35 MARKS]

Explain the Merge Sort algorithmic paradigm. State its recurrence relation in divide-and-conquer form and derive the tight asymptotic time complexity bound using the Master Theorem. Demonstrate the split and merge flow step-by-step on the 8-element array: **[38, 27, 43, 3, 9, 82, 10, 19]**.

> **Evaluation Rubric Key:**
> - Recurrence formulation: T(n) = 2T(n/2) + Θ(n) [10 Marks]
> - Master Theorem Case 2 application: a=2, b=2 ⇒ T(n) = Θ(n log n) [10 Marks]
> - Step-by-step 2-way split & recombine trace on array [10 Marks]
> - Space complexity auxiliary buffer proof: O(n) [5 Marks]

PART B — (2 × 35/30 = 65 Marks)

Question 2. [Self-Balancing Binary Search Trees] [35 MARKS]

Define an AVL Tree and explain how the balance factor is maintained within {-1, 0, +1}. Construct an AVL tree by sequentially inserting the key sequence: **[15, 20, 24, 10, 13, 7, 30, 36, 25]**. Illustrate the balance factor after every insertion and specify whether Single (LL, RR) or Double (LR, RL) rotations are triggered.

> **Evaluation Rubric Key:**
> - Definition: BF = Height(Left_Subtree) - Height(Right_Subtree) [10 Marks]
> - RR rotation executed at Node 15 upon inserting 24 [10 Marks]
> - LR double rotation executed at Node 15 upon inserting 10 and 13 [10 Marks]
> - Final verified balanced tree structure [5 Marks]

Question 3. [Single-Source Shortest Paths & Graph Algorithms] [30 MARKS]

Formulate Dijkstra's algorithm for finding the shortest path from a single source vertex in a directed weighted graph. State the critical constraint on edge weights. Explain how a Min-

Priority Queue is employed during edge relaxation and compute the overall worst-case running time complexity.

> **Evaluation Rubric Key:**
> • Algorithm formulation & non-negative weight prerequisite [10 Marks]
> • Relaxation step: if dist[u] + w(u,v) < dist[v] then dist[v] = dist[u] + w(u,v) [10 Marks]
> • Min-Heap complexity analysis: O((V + E) log V) with auxiliary space O(V) [10 Marks]